

Assessing Eye Tracking Accuracy in Real-World AR Locomotion Scenarios

Category: Research

ABSTRACT

Eye tracking is an increasingly popular method for interacting with augmented reality (AR) and measuring user attention. However, implementing and evaluating eye tracking across multiple platforms and use cases can be challenging due to the lack of standardized metrics and measurements. Additionally, existing calibration methods and accuracy measurements often do not account for the common scenarios of walking and scanning in mobile AR settings. To test and compare different eye tracking devices on various AR tasks and metrics, we conducted three user studies evaluating eye tracking on three mixed reality (MR) headsets: the Magic Leap One, the HoloLens 2, and the Meta Quest Pro. These user studies incorporated tasks that tested eye tracking performance while stationary and while walking. Additionally, we performed post-hoc recalibration of the eye gaze data using the tasks in our test suite in order to study the impact of different calibration tasks on accuracy and compare them with the inbuilt calibration provided by the head-mounted display (HMD)s. Our results revealed that the degree to which locomotion influenced eye tracking performance depended on the headset, with the HoloLens 2 displaying the greatest decrease in accuracy during locomotion. We speculate that this may be due to the way the headsets fit on the user's head.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality; Human-centered computing—Human computer interaction (HCI)—HCI design and evaluation methods—User studies; Human-centered computing—Ubiquitous and mobile computing—Ubiquitous and mobile devices—Mobile devices;

1 INTRODUCTION

Eye tracking is a useful tool that has gained popularity in recent years. Its many applications, from diagnostics (gathering data on a user's visual attention) to interaction (allowing users to use their gaze as a cursor) [11], make it a useful means of understanding user behavior and enhancing human-computer interaction. Recent advancements in eye-tracking technology have made it more accessible and affordable for a wider range of applications [15]. As such, it is important that developers and users understand the capabilities and limitations of their eye tracking devices during use.

With the rise of AR HMDs such as the Magic Leap, Meta Quest Pro, HoloLens 2, and the recently-announced Apple Vision Pro, eye trackers have been integrated into various platforms and use cases. However, implementing eye-tracking technology on multiple platforms and standardizing their performance is a difficult and complex problem. To address this challenge, specific metrics, such as accuracy and precision, are often selected to evaluate the performance of eye-tracking systems [10, 15, 21]. Although existing work has used these metrics to examine the accuracy of eye trackers while stationary, there has been little research into how locomotion affects eye tracking. Even the announcement for the Apple Vision Pro, which integrates eye tracking and interaction more than any previous AR headset, did not contain any examples of eye tracking used during locomotion [4]. It is clear that the effects of locomotion on eye tracking performance are underexplored and important to understand as AR use continues to grow.

Furthermore, current eye tracking device calibration methods rely on users keeping their heads stationary, without incorporating any head or body movement. It is unclear whether these existing

eye tracking calibration methods are effective when a user intends to use eye tracking while walking. Therefore, it is important to investigate how effective existing eye tracking calibration methods are and potentially find new calibration procedures to incorporate locomotion.

In order to meet these needs, we designed and ran three user studies to assess eye tracking performance in various AR scenarios that involve movement and locomotion. The studies were run on three MR headsets: the Magic Leap One, the Meta Quest Pro, and the HoloLens 2. The user studies consist of a comprehensive range of tasks that simulate various scenarios encountered in AR applications in different conditions: with the user's head constrained to a chin rest, with the user's body constrained through sitting, and with the user free to move and walk. In addition, these tasks present a stimulus in two reference frames: world or screen stabilized. The procedure followed in this user study provides a standardized approach for evaluating both user performance and device performance within these specific scenarios.

Our goal in running these user studies was to assess the spatial accuracy and precision of the gaze signal captured by the integrated eye tracker of the Magic Leap One, Meta Quest Pro, and HoloLens 2 headsets. We analyzed eye tracking performance in different movement conditions and investigated possible recalibration procedures to improve eye tracking accuracy during locomotion. Our analysis revealed that 1) eye tracking accuracy tended to decrease as head and body motion increased, 2) the different headsets yielded significantly different results on several tasks, and 3) introducing controlled head motion to a recalibration task resulted in better performance for eye tracking during locomotion on the HoloLens 2.

To ensure the reproducibility of our analysis, we will make the datasets from both studies as well as the source code of the test suite publicly accessible to the research community.

2 BACKGROUND AND RELATED WORK

Here we present prior work on stationary eye tracking evaluation, existing calibration methods, and eye tracking during locomotion.

2.1 Eye Tracking Evaluation

Existing work has evaluated the accuracy of eye trackers, but only in a stationary lab setting. Weigle et al. present a protocol to determine the accuracy of an eye tracking device, which consists of users watching the top and bottom edges of an animated bouncing stimulus on a screen [31]. Ehinger et al. present a more rigorous protocol for determining eye tracking accuracy, which consists of ten tasks that present stimuli on a screen and measure accuracy during smooth pursuit, microsaccades, head movements, and other conditions [16]. Aziz and Komogortsev evaluate eye tracking on a HoloLens 2 device using a random saccades task, in which participants fixate on a stimulus presented at random points in the visual field in front of them [10]. All of these protocols present stimuli on a 2D screen or field of view (FOV), rather than a 3D space, and none of them involve locomotion.

There has also been work into creating generalized packages to evaluate eye trackers. Adhanom et al. present GazeMetrics, a Unity package that includes a calibration procedure to evaluate the accuracy of several supported eye trackers, including the HTC VIVE eye tracker and Pupil Labs add-on eye tracker [8]. Kapp et al. present the Augmented Reality Eye Tracking Toolkit (ARETT), an

eye tracking Unity package for the HoloLens 2 that records eye tracking data and includes an R package for data analysis [21].

In spite of the growing prevalence of head-mounted eye trackers, there is limited work on evaluating eye tracking while moving. The researchers behind ARETT showed that eye tracking accuracy for the HoloLens 2 is lower when a user is walking than when they are standing still [21], but the conditions with which they tested eye tracking while walking were limited to a screen-stabilized saccades procedure. Hausamann et al. examine simultaneous head and eye tracking in a moving setting, but focus on the accuracy of the head and eye tracking data relative to each other, rather than on eye tracking specifically [18].

2.2 Eye Tracking Calibration

Eye tracking calibration involves finding the mapping between the output of an eye tracker and a user's gaze point. There is existing work on finding good calibration methods and functions. Harezlak et al. describe a standard calibration procedure in which users look at defined points on a screen, and a polynomial function is used to map their eye tracking output to those points [17]. Kasprowski et al. find that linear and quadratic functions generally perform better than more complex functions when calibrating eye trackers [22]. Mardanbegi et al. suggest that errors may arise in accuracy when viewed content is at a different depth than the calibration plane [25]. Although existing work thoroughly examines different aspects of the ideal calibration procedure, there has not been any work in finding or generalizing a calibration procedure for eye tracking during locomotion in a 3D environment.

2.3 Eye Tracking during Locomotion

Prior work exists that utilizes eye tracking during locomotion in order to investigate scientific phenomena. Kiefer et al. examine the applications of eye tracking in spatial research and note that eye tracking may provide a valuable tool for researchers looking to investigate how users navigate during locomotion [23]. De Winter et al. demonstrate this by using eye tracking during walking to examine where people focus their attention when walking through a parking garage [13].

Ophthalmologists routinely make use of eye tracking during locomotion to evaluate patients' visual health. Vargas-Martin et al. use mobile eye tracking on patients with severe peripheral visual field construction in order to make recommendations for head-mounted mobility aids [30]. Stuart et al. present an algorithm for detecting saccades during locomotion on the Dikablis eye tracker and test this algorithm on healthy users and users with Parkinson's disease [28]. Vanegas-Arroyave et al. use eye tracking during locomotion to investigate where patients with Parkinson's focus their attention during walking [29].

While eye tracking is increasingly applied in mobile settings, thus coupling eye tracking with ambulatory locomotion, there is still a paucity of studies on how currently available eye trackers are affected by locomotion, especially in the Mixed Reality domain. This work attempts to start closing this knowledge gap and provide evaluation and calibration tools for AR/VR researchers interested in eye tracking while walking.

3 SYSTEM AND IMPLEMENTATION

In this section, we introduce the eye tracking hardware capabilities of the Magic Leap One, Meta Quest Pro, and HoloLens 2 headsets, as well as the development of the user study test suite.

3.1 Eye Tracking Hardware

The Magic Leap One utilizes one camera per eye for eye tracking [6]. According to Caruso et al, eye gaze information is provided at 60 frames per second using the pupil center and corneal reflections (PCCR) [12,32]. Based on this information, we anticipate a precision

of up to 1° [20]. Although there is limited research on evaluating the accuracy of the Magic Leap device, we can refer to the spatial accuracy of 1.5° for HoloLens 2 [5], a comparable headset that also utilizes PCCR, as a reference point. The MLEyes API offers the gaze position as a 3D point in space, gaze confidence, and a single-direction gaze vector. Additionally, it provides individual gaze confidence, blinking state, eye position, gaze position, and pupil size for both eyes.

The Meta Quest Pro has a visible field of vision of 106° horizontally and 95.57° vertically. It consists of 5 total infrared sensors capable of a 120° field of view each, 2 of which are used for eye tracking [1, 2]. The headset has a refresh rate of 90Hz [19]. Eye tracking is enabled through the Movement SDK for Unity, and is available via the OVREyeGaze MonoBehaviour component. An OVREyeGaze script can be attached to a GameObject to represent either the left or right eye, updating the GameObject's position and orientation based on the eye movement. Thus, we can get the positions and forward vectors of the left and right eyes. The SDK does not report a vergence point, blinks, pupil location, or pupil size.

The HoloLens 2 integrated eye tracker consists of two infrared cameras, one for each eye, [7] that use corneal reflections to determine a gaze direction. The eye tracking accuracy is reported to be within 1.5 degrees with a sampling rate of 30 Hz [5]. The HoloLens 2 eye tracking API reports gaze as a single direction vector and a gaze origin, without reporting individual eye gaze vectors, a vergence point, blinks, or pupil location.

3.2 User Study Development

Initially, we planned to run these user studies exclusively on optical see-through headsets, such as the Magic Leap One and HoloLens 2. However, we realized that the video passthrough mode of the Meta Quest Pro also provided opportunities for AR development on this and other headsets. As a result, we decided to include the Meta Quest Pro in our suite of user studies, in spite of it being a virtual reality (VR) headset rather than AR.

The set of tasks in the user study was developed for the Magic Leap headset using the Magic Leap Lumin SDK 0.26, which officially supports Unity editor version 2020.3. We utilized Unity 2020 LTS along with Magic Leap XR Plugin version 6.2.2. For the HoloLens 2, EyeTTS was built using Unity 2019.4.30f1 and the Mixed Reality Toolkit version 2.7.0 with the Windows XR plugin. The package consists of a single Unity scene implemented with Unity World Locking Tools enabled. The Meta Quest Pro EyeTTS was developed using Unity 2021.3.15f1 and the Unity Oculus Integration Package v55.

Our user study included tasks that required participants to maintain their gaze on stimuli, which is common in eye gaze studies [14, 26]. The stimuli could be stationary or moving relative to the user and could be presented on a world-stabilized or screen-stabilized reference frame. This allowed for data collection from conditions representative of various situations in AR.

Each of the tasks involved a hold-and-go movement pattern, in which the stimuli followed a seemingly random but predetermined path. These alternating phases were labeled appropriately during data collection to segregate static and dynamic stimulus data points. During the hold phase, the gaze data captured corresponded to fixations, representing static elements within the augmented environment. Conversely, in the go phase, the stimulus observed represented animated and moving elements. The gaze data in the go phase could encompass fixations, smooth pursuit, or saccades, depending on the specific conditions of the task.

We designed the stimuli movement patterns to cover the entire FOV of the user. A frame was placed around the area where the stimulus moved, which was ensured to be within the FOV by making positional and orientational adjustments before starting the task. This frame was not visible during the task.

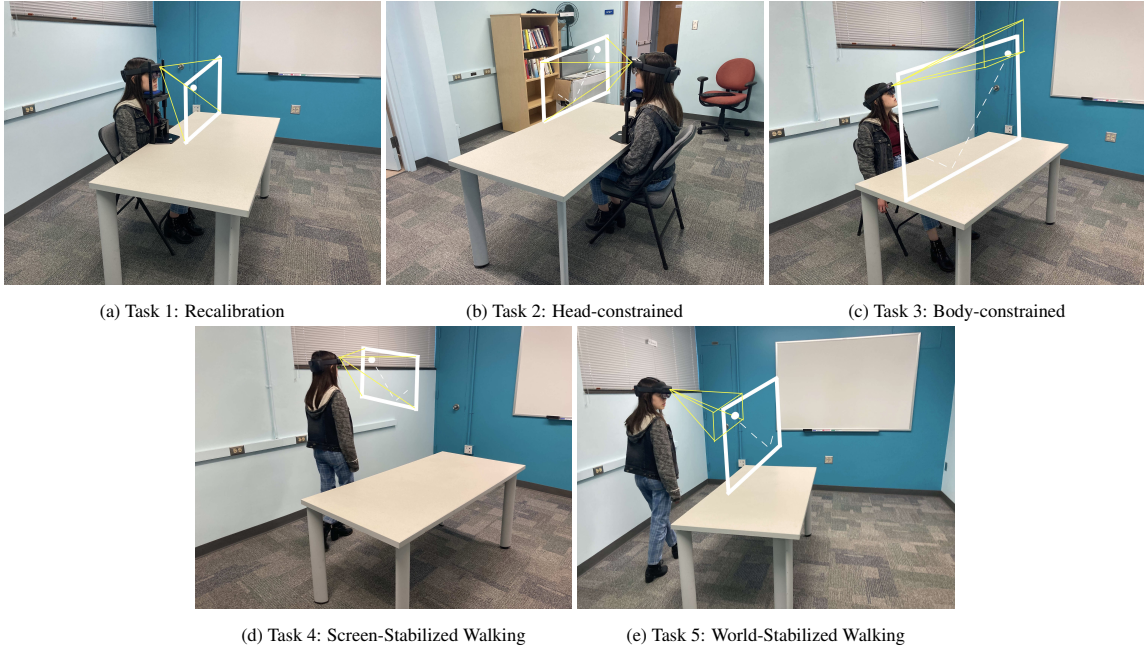


Figure 1: The tasks performed in our user study. AR content is colored in white, while yellow lines approximate the FOV of the headset. The users did not see the frame or path (dashed line) of the target once the experiment began.

4 USER STUDY DESIGN

Three user studies were sequentially carried out to investigate eye tracking performance on different MR headsets. The initial user study ($n=36$) was conducted utilizing the Magic Leap headset. A subsequent user study ($n=29$) was carried out on the Meta Quest Pro headset. The final user study ($n=54$) was performed using the HoloLens 2 headset, with an additional task added based on feedback and results from previous studies.

4.1 Participants

The first user study on the Magic Leap included a total of 36 university students, with an average age of 22.75 and an equal number of participants identifying as male and female. Out of the total participants, 26 had prior experience with VR or AR headsets and had previously worn a headset. The experiment consisted of six tasks, and each participant performed each task once.

The second user study on the Quest Pro consisted of 29 total participants ranging from ages 19 to 29 years, with an average age of 21.9 years. Of the participants, 15 identified as male, 12 identified as female, and 2 identified as non-binary. The participants were recruited from a university campus. 19 of the participants had previous experience with AR, and 27 had previous experience with VR. 12 participants who usually wore glasses removed them for this study, and 2 participants wore contact lenses while participating. None of the participants wore glasses during the study.

The third user study on the HoloLens 2 included a total of 54 participants, of which 29 identified as female and 25 identified as male. The participants were recruited from a university campus and had an average age of 22.4 years. Of the participants, 34 had previous experience with AR.

4.2 Experiment Procedure

To begin, each participant was asked to read and sign a consent form prior to beginning the experiment. Upon signing the consent form, each participant then underwent the in-built eye calibration procedure built onto the device while keeping their head stationary by resting it on a Tobii Chin Rest. Following the calibration, each

participant completed a pre-study survey/questionnaire, providing demographic information and assessing their familiarity with MR. Subsequently, participants engaged in the tasks under researcher instruction and supervision.

4.3 Tasks

Each user study had five tasks in common to ensure comparability. To reduce order effects, predefined task orders were generated using the Latin Square method, which is a balanced and counterbalanced design technique. Furthermore, most tasks adhered to the hold-and-go movement pattern for the target, as described in Sect. 3.2.

We developed tasks for our user study based on various frames of reference in typical mobile AR applications. We refined this concept by considering precise spatial and orientational relations between the user, their head, and virtual objects. This led to a taxonomy of AR experiences in different frames of reference, forming the initial pool of tasks. We then filtered these based on their prevalence in AR applications, ensuring relevance for our user study. To further refine our test suite, we compared occurrences in popular AR and VR applications and games, eliminating any duplicates. We initially identified 144 different usage categories that we pruned down to 5.

All tasks featured screen-stabilized or world-stabilized content. In user study 1 on the Magic Leap, world-stabilized content was anchored to the real world with images detected by the Magic Leap XR Plugin. For user studies 2 and 3 on the HoloLens 2 and Meta Quest Pro, the world-stabilized content was manually anchored to the real world by the researcher prior to the experiment.

Several tasks involved participants walking in circles around a table. In order to minimize directional effects in these walking tasks, half of the participants were asked to walk clockwise around the table, while the other half were asked to walk counter-clockwise.

4.3.1 Task 1: Head-Constrained Static

The static head-constrained task aimed to simulate the eye calibration procedure found in the headsets' built-in systems. Participants were asked to rest their heads on a Tobii Chin Rest in order to keep their heads from moving during the task. During the task, a tracking



Figure 2: The hallway task, which was only executed on the HoloLens 2 user study

stimulus appeared at random points in front of the participant, within their entire FOV. The stimulus did not move in between points, and it remained at each point for 2.5 seconds. The AR content was screen-stabilized and programmed to appear 1.25m-1.5m away from the participant’s eyes, as recommended by [3]. Participants were asked to look at the stimulus as soon as it appeared without moving their head.

4.3.2 Task 2: Head-Constrained Moving

Task 2 was a head-constrained moving task with a similar setup to the head-constrained static task. Participants were asked to rest their heads on a Tobii Chin Rest to minimize head motion. The tracking stimulus moved on preprogrammed paths within a screen-stabilized plane located 1.25m in front of the user. The stimulus took 5 seconds to complete a path and stayed in place for 1.5 seconds between movements. The speed of the stimulus during the moving phase consistently measured below 40°/sec, remaining within the smooth pursuit limit of 60°/sec [27]. Therefore, we classify this screen-stabilized task as a smooth pursuit task, which is relevant to our precision calculations in Sect. 5.3.

4.3.3 Task 3: Body-Constrained

Task 3 was a body-constrained fixation task, meant to emulate a situation in which a user is stationary in augmented space while observing content that is world-stabilized. Participants could move their heads freely, but were asked to stay seated throughout the task. Furthermore, we requested the participant to use head movements to always maintain the target in the center of their FOV. In this task, the tracking stimulus moved on preprogrammed paths around a world-stabilized plane located in front of the participant. The paths were designed to be sufficiently long enough to require participants to turn their heads in order to observe the stimulus.

4.3.4 Task 4: Screen-Stabilized Walking

Task 4 was an unconstrained task in which participants were asked to walk in circles around a table. The tracking stimulus moved in pre-programmed paths around a screen-stabilized space located 1.25m in front of the participant. Similar to task 2, the head-constrained task, this task was categorized as a smooth pursuit task. However, unlike task 2, it presented the additional challenge of incorporating locomotion while performing an eye tracking task.

4.3.5 Task 5: World-Stabilized Walking

Task 5 was an unconstrained task in which participants walked in circles around a table, as shown in Fig. 1e. Unlike in task 4, the screen-stabilized walking task, the tracking stimulus moved in

preprogrammed paths around a world-stabilized space located over the table. This forced the participant to turn their head to look at the stimulus while walking.

4.3.6 HoloLens 2 Task: Hallway

In the course of running these user studies, we received participant feedback that walking in tight circles during the walking tasks felt very unnatural, even making some participants nauseous. As a result, we decided to add a sixth task to the HoloLens 2 study, with the goal of creating a more natural walking experience in augmented reality. In this hallway task, participants were asked to walk down a hallway and follow the tracking stimulus as it moved back and forth across the hallway in front of them, as shown in Fig. 2. The participants were not provided with specific instructions regarding walking speed or distance from the stimulus, in order to encourage natural walking.

We assess the effectiveness of our recalibration efforts on the hallway task in section 5.4. Given that this task closely mirrors typical AR scenarios involving walking, any enhancements here would be significant.

4.4 Data Analysis

In this section, we go over the calculations done in our data analysis.

4.4.1 Data Preprocessing

When visualizing the data recorded by the HoloLens 2, a latency between the reported head position and the reported eye position was observed. We minimized the latency between the head position and the eye position by shifting the eye position and corresponding eye gaze data backward until the mean euclidean distance between head position and eye position was minimized. This shift was limited to 1.2 seconds. This latency was not observed for data collected on the Magic Leap One or Meta Quest Pro, so this correction was not performed for those user studies.

To correctly analyze data from task 1, the head-constrained static task, we needed to identify periods of stable fixation and eliminate errors due to saccadic movements. The initial period in the fixations consists of instability due to non-fixational movements [10]. Aziz and Komogortsev [10] determined that by excluding the initial 233 ms of fixations, the largest contiguous collection window could be obtained with minimal errors attributed to such instability. We opted to discard the initial 0.5 seconds of data collected each time the tracking sphere changed positions or settled on a new position after movement, while retaining the subsequent 1.5 seconds of data.

We used the full 90 (1.5s * 60Hz) samples from the static segments and randomly selected 90 samples from each of the dynamic segments to arrive at our comparison between static and moving stimuli (Fig. 4).

4.4.2 Recalibration

Although each participant completed the Magic Leap One, HoloLens 2, or Meta Quest Pro’s built-in eye calibration, we wanted to see whether we could improve on these results with further recalibration on the other tasks. To that end, we used linear regression:

$$x' = A_x x + B_x y + C_x, \quad y' = A_y x + B_y y + C_y$$

The coefficients A, B, and C were calculated using the data from whichever task was used for recalibration, such as task 1, and the coefficients were applied to data from other tasks.

4.4.3 Cosine Error

In order to calculate cosine error, we used three-dimensional gaze and target vectors. The HoloLens 2 eye tracking API reported eye gaze direction as a 3-dimensional vector. For the Magic Leap, which reported a gaze point rather than a direction, the direction vector was calculated using that gaze point and reported head position.

Device	Overall	Transition	Fixation	Smooth Pursuit
Magic Leap	1.5038°	3.6378°	0.4193°	0.4543°
HoloLens 2	0.665°	2.2699°	0.1028°	0.1428°
Quest Pro	1.426°	2.895°	0.6613°	0.7220°

Table 1: Precision Values for the Magic Leap and HoloLens 2

The Quest Pro reported a forward vector and position for each eye. To obtain a singular gaze point, we estimated the vergence point by finding the midpoint of the shortest segment perpendicular to both individual gaze rays, as done in [9]. We then converted the vergence point into a three-dimensional gaze vector by forming a point between the head position and the vergence point. The gaze vector v and the target vector u were used to calculate cosine error using the following formula:

$$\theta = \arccos \frac{\vec{u} \cdot \vec{v}}{||u|| ||v||}$$

Cosine errors for each sample were averaged by participant, and each participant's average cosine error was averaged again to calculate the total average cosine error.

4.4.4 Euclidean Error

In order to convert the 3D gaze direction and target vectors ($v = (v_x, v_y, v_z)$) into 2D screen coordinates, we used the following formulas:

$$x = \arctan\left(\frac{v_x}{v_z}\right), \quad y = \arctan\left(\frac{v_y}{v_z}\right)$$

We then used the 2D screen coordinates to calculate Euclidean error:

$$\theta = \sqrt{(x_{target} - x_{gaze})^2 + (y_{target} - y_{gaze})^2}$$

Euclidean errors for each sample were averaged by participant, and each participant's average error was averaged again to calculate the total average Euclidean error.

4.4.5 Spatial Precision

Spatial precision for transition periods, fixations, and smooth pursuit was calculated using the data from tasks 1 and 2. The following formula was used:

$$\theta_{RMS} = \sqrt{\frac{1}{n} \sum_{i=1}^n (\theta_i)^2}$$

where θ_i is the Euclidean distance between consecutive samples. n represents the number of samples in one fixation or period of movement. Precision for the transition period between fixations was calculated using the first 0.5 seconds after each target movement in task 1. Precision for fixation was calculated using the remainder of the time the target was stationary for each position in task 1. Finally, precision for smooth pursuit was calculated using the periods when the target was moving in task 2. Each period of target movement or fixation was considered to be n samples, and the precision value was averaged over one participant. Average precision values for each participant were then averaged to find an overall precision value.

5 RESULTS

Results for all studies are presented here.

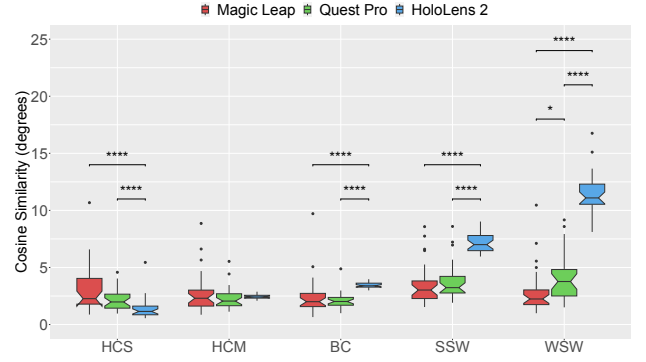


Figure 3: Cosine similarity per task for the Magic Leap, Meta Quest Pro, and HoloLens 2. Notches show 95% confidence interval, * denotes $0.05 \geq p \geq 0.01$, ** denotes $0.01 \geq p \geq 0.001$, *** denotes $0.001 \geq p \geq 0.0001$, and **** denotes $0.0001 \geq p \geq 0$.

5.1 Cosine Error

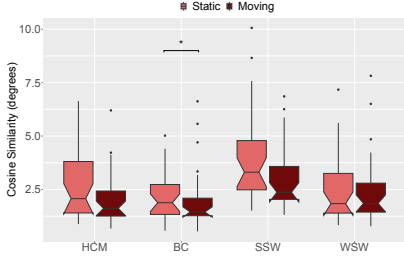
Fig. 3 shows a graph of cosine error split by task for all three devices. Due to the different numbers of participants in each user study, the data from 29 random participants was selected from the Magic Leap One and HoloLens 2 studies in order to match the quantity of data available from the Meta Quest Pro and allow for a better comparison of the three studies. An ART ANOVA with independent variables task and device revealed a significant difference in cosine error between tasks ($F = 341, p < 2.22 \times 10^{-16}$), a significant difference in cosine error between devices ($F = 49, p < 8.77 \times 10^{-15}$), and an interaction between task and device ($F = 147, p < 2.22 \times 10^{-16}$). For the Magic Leap One and Meta Quest Pro, a Bonferroni-corrected Mann-Whitney U posthoc test reveals a difference in cosine error in the world-stabilized walking task ($p = 0.0024$). For the Magic Leap One and HoloLens 2, the same posthoc test reveals a difference in cosine error in the head-constrained static task ($p = 1.94 \times 10^{-6}$), the body-constrained task ($p = 1.01 \times 10^{-6}$), the screen-stabilized walking task ($p = 1.32 \times 10^{-7}$), and the world-stabilized walking task ($p = 4.02 \times 10^{-10}$). For the Meta Quest Pro and HoloLens 2, the posthoc test reveals a difference in cosine error in the same tasks: the head-constrained static task ($p = 7.5 \times 10^{-5}$), the body-constrained task ($p = 3.6 \times 10^{-9}$), the screen-stabilized walking task ($p = 6.48 \times 10^{-8}$), and the world-stabilized walking task ($p = 2.4 \times 10^{-10}$).

5.2 Stimulus Movement

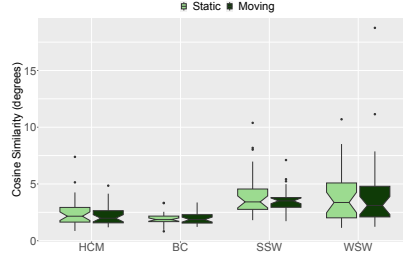
In order to evaluate accuracy for different target movements, we separated the data according to when the tracking stimulus was moving between points and when the tracking stimulus was static. Task 1, the head-constrained static task, only contained static data and was excluded from this analysis. As outlined in section 4.4.1, we also sampled the data from when the tracking stimulus was moving in order to have equal amounts of data from when the stimulus was moving and static.

Fig. 4 shows graphs of cosine error for the static and moving stimulus for all user studies. For the Magic Leap One, an ART ANOVA with independent variables task and movement revealed a significant effect of movement on cosine error ($F = 11.63, p = 7.89 \times 10^{-4}$). A Bonferroni-corrected Mann-Whitney U test revealed a significant effect of stimulus movement on the screen-stabilized walking task ($p = 0.0416$). Fig. 4 suggests that cosine error is higher when the target is static for this task.

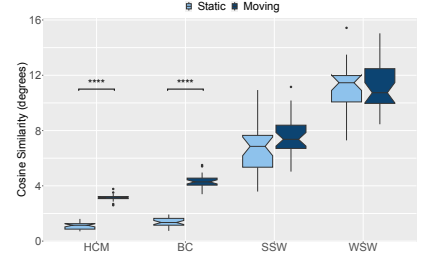
For the Meta Quest Pro, an ART ANOVA with independent variables task and movement does not reveal a significant effect of target movement on cosine error.



(a) Cosine error for the Magic Leap, separated by stimulus movement.



(b) Cosine error for the Meta Quest Pro, separated by stimulus movement.



(c) Cosine error for the HoloLens 2, separated by stimulus movement.

Figure 4: Cosine error graphs for stimulus movement for the Magic Leap, Meta Quest Pro, and HoloLens 2. Notches show 95% confidence interval, * denotes $0.05 \geq p \geq 0.01$, ** denotes $0.01 \geq p \geq 0.001$, *** denotes $0.001 \geq p \geq 0.0001$, and **** denotes $0.0001 \geq p \geq 0$.

For the HoloLens 2, an ART ANOVA with independent variables task and movement reveals a significant effect of target movement on cosine error ($F = 154.95, p < 2.22 \times 10^{-16}$), and an interaction between task and movement ($F = 37.76, p < 2.22 \times 10^{-16}$). A Bonferroni-corrected Mann-Whitney U test revealed a significant difference in error when the target is moving or static for the head-constrained moving task ($p = 6.51 \times 10^{-11}$), the body-constrained task ($p = 6.51 \times 10^{-11}$). Fig. 4 suggests that cosine error is higher when the target is moving for both tasks.

5.3 Spatial Precision

Table 1 shows precision values for the Magic Leap, Meta Quest Pro, and HoloLens 2 in various conditions taken from both head-constrained tasks. As expected, all headsets exhibit a greater precision during fixation and smooth pursuit than during the transition period between stimulus locations.

5.4 HoloLens 2 Recalibration

Post-hoc recalibration was performed on every task in order to find a potential recalibration method suited for eye tracking during locomotion. Results from the Magic Leap and Meta Quest Pro was not greatly affected by recalibration. As a result, only recalibration results for the HoloLens 2 are presented.

In order for a task to be a suitable candidate for recalibration, it must both result in improvements and be easy to perform in any situation prior to AR use. Specifically, we are interested in improvements to eye tracking during locomotion, so we decided to prioritize recalibrations that resulted in improvements to the walking tasks over recalibrations that resulted in improvements to the head-constrained or body-constrained tasks. Furthermore, even if recalibrating on a walking task results in improvements to other walking tasks, a walking recalibration task would require a specialized environment for every user and would be impractical for recalibration. Out of the tasks in our user study, this leaves the head-constrained static, head-constrained moving, and body-constrained tasks as suitable tasks for recalibration. Looking at the results of those recalibrations, we choose to focus on the body-constrained task for the bulk of our analysis, as it brought about the most consistent improvement in eye tracking performance while walking in the HoloLens 2.

Fig. 5 shows a graph of Euclidean error with device calibration and recalibration using the body-constrained task for both the HoloLens 2 study. This graph includes all data collected on the HoloLens 2, which includes all 54 participants. In addition, this graph includes the hallway task, which was only completed on the HoloLens 2 and is a more representative walking task than the other walking tasks. As shown in the graph, device-calibrated Euclidean error for the hallway task is lower than the error on either walking task, but higher than the error on non-walking tasks.

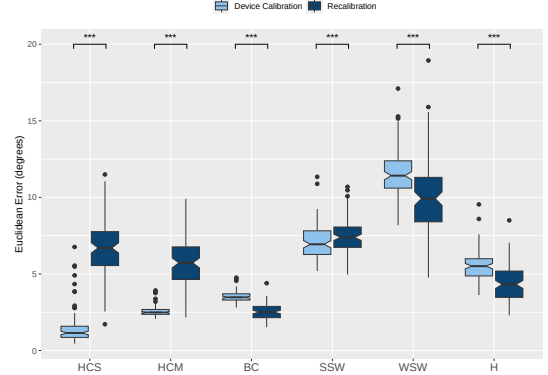


Figure 5: Euclidean error for all tasks of the HoloLens 2, separated by recalibration on the body-constrained task. Notches show 95% confidence interval, *** denote $p < 0.001$.

An ART ANOVA with independent variables task and calibration type reveals a significant effect of recalibration on Euclidean error ($F = 234, p < 2.22 \times 10^{-16}$), and an interaction between task and recalibration ($F = 327, p < 2.22 \times 10^{-16}$). A Bonferroni-corrected Mann-Whitney U test revealed a significant difference between device-calibrated and recalibrated error for all tasks ($p_{Task1} = 8.46 \times 10^{-35}, p_{Task2} = 2.99 \times 10^{-34}, p_{Task3} = 1 \times 10^{-29}, p_{Task4} = 5.56 \times 10^{-3}, p_{Task5} = 4.56 \times 10^{-9}, p_{Task6} = 2.51 \times 10^{-11}$). Figure 5 suggests that recalibration may have resulted in a higher Euclidean error for the head-constrained static, head-constrained moving, and screen-stabilized walking tasks, but a lower Euclidean error for all other tasks.

6 DISCUSSION

Our results include both, intrinsic differences between the HMDs, as well as general trends and insights that could improve eye tracking performance for AR walking.

The Magic Leap data exhibited a limited loss of accuracy due to locomotion or head motion. The only pairwise comparisons of tasks that showed any statistically significant differences were the head-constrained task vs. the body-stabilized walking task, the body-constrained task vs. the screen-stabilized walking task, and the body-constrained task vs. the body-stabilized walking task. This suggests that the Magic Leap may encounter a loss of eye tracking accuracy with the addition of ambulatory locomotion, but not simply with the addition of head motion in the body-constrained case.

For the Quest Pro, there was a significantly significant difference between every pairing of a stationary task (HCS, HCM, BC) and a

walking task (SSW, WSW). The results suggest a significant difference between tasks that are stationary, and those that allow free body rotation or movement. This implies that the Quest Pro eye tracking performs more accurately in stationary or seated settings compared to any types of head motion.

Despite every participant performing a fit adjustment prior to the study, it was observed that some participants had to adjust the headset throughout the study. Moreover, during the walking tasks, some participants reported that the target was unstable and “vibrated.” This suggests that the headset is sensitive to pronounced head movement and the vibrations from walking, implying that the headset fit and security could be improved upon.

Continuing this trend, the HoloLens 2 has an even higher error than either of the other two headsets and displays a trend in which head-constrained tasks are the most accurate, followed by the body constrained task, then walking tasks, with rapidly increasing eye tracking error. Latency between the head position and the eye position in all walking tasks may have resulted in the lack of accuracy for the walking tasks. Although we attempted to correct this latency by shifting all walking gaze data up to 1.2 seconds post-hoc, following the advice by [24], it is possible that this correction was not entirely effective in reducing resulting errors. This latency may have been exacerbated by the HoloLens 2’s lower eye tracking sampling rate of 30 Hz (through Unity), which is lower than the 60 Hz eye tracking sampling rate of the Magic Leap.

Among the walking tasks, the screen-stabilized walking task had a much higher error than the world-stabilized walking task. Several participants reported some dizziness and nausea during the screen-stabilized walking task. Therefore, it is possible that the higher errors in the screen-stabilized walking task are due to the physical difficulty of the task. The numerous changes in head orientation in this task could also have caused the eye tracker to be less accurate.

The only task for which it seemed that the HoloLens 2 exhibited better eye tracking performance than the Magic Leap and Meta Quest Pro was the head-constrained static task. This can potentially be explained by the larger distance between the display and the user’s eyes on the HoloLens 2, which may have allowed the eye tracking cameras to better capture eye gazes at the periphery of the display. Differences in the built-in calibration procedure may also have an effect here.

The differences between the accuracy of the HoloLens 2 compared to the Magic Leap One and Meta Quest Pro may largely be explained by the fit of the headsets. The Magic Leap One rests on the nose, forming a secure fit around a user’s head, and is the only headset having its compute unit separated from the headset, which contributes to its secure fit. The Meta Quest Pro rests on a band fitted around the forehead, with the display fixed in place. In contrast, the HoloLens 2 has a visor housing the display that can be flipped up, which can result in a lack of stability between the display and the face. This potentially results in more headset movement relative to the head and lower accuracy for tasks when head and body motion increase. The lack of stability of the HoloLens 2 on a user’s head could explain why the eye tracker is significantly more accurate when the head is kept still, whereas the Magic Leap One and Meta Quest Pro headsets show more consistent eye tracking accuracy throughout.

The only task for which the Meta Quest Pro and Magic Leap have significantly differing eye tracking performance is the world-stabilized walking task, in which the Meta Quest Pro has a lower accuracy than the Magic Leap One. This may be due to the Meta Quest Pro using a video-passthrough display instead of an optical-see-through display, which may affect task performance. Seeing the world through an imperfect video feed while navigating around a central table that is to be kept in focus is more taxing to the human visual system than using an optical-see-through display.

Regarding moving stimuli, the Meta Quest Pro did not exhibit

any differences in accuracy due to stimuli movement. In contrast, the HoloLens 2 data exhibited a clear trend of stimulus movement resulting in lower accuracy compared to the static case for some tasks. The Magic Leap One data appears to maintain a high accuracy when the target is moving. It is possible that the higher sampling rate of the Magic Leap allows it to handle stimulus and gaze movement more effectively than the HoloLens 2.

We decided to only use the head-constrained static and head-constrained moving tasks for precision calculations because they match the existing literature most closely when it comes to this type of analysis. Unsurprisingly, transition periods, which are affected more by saccades, produce less precise data than smooth pursuit or fixation periods for all headsets. However, it is worth noting that the eye tracking APIs in our user studies do not provide information that could be used to help distinguish between a fixation or transition point. We therefore approximated the transition period as happening up to 0.5 seconds after the target shifted, which is conservative compared to existing literature (e.g. 233 ms in [10]).

Overall, the HoloLens 2 results differed the most from those of the other two headsets, with significantly higher errors, especially in walking tasks. This may be due to the difference in display quality between the HoloLens 2 and the other headsets, which sometimes made the tracking stimulus difficult to see during the task. Although the hallway task was more accurate than either of the other two walking tasks, it was still less accurate than any of the tasks without locomotion.

The overall lower accuracy on the HoloLens 2 may explain the relative effectiveness of recalibration compared to the other two headsets. When it comes to recalibration, no recalibration task improved performance on every single task. Instead, it was often the case that recalibration on a task that improved conditions during locomotion, such as the body-constrained task, resulted in worse performance during the head-constrained tasks. This may suggest that it is important to consider the end use case of a headset when choosing a recalibration task, and a task with more head motion, such as the body-constrained task, may be an appropriate choice for recalibration for locomotion. The improvement on the hallway task, when recalibrated with data from the body-constrained task shows, promise that a recalibration task incorporating head motion may be effective in recalibrating for locomotion.

7 LIMITATIONS AND FUTURE WORK

Creating comparative user studies for three distinct headsets – the Magic Leap, HoloLens 2, and Meta Quest Pro – was complicated due to their fundamentally different architectures, designs, capabilities, and software development kits (SDKs). Each of these devices operates on distinct technological frameworks, resulting in variations in how they handle augmented reality (AR) interactions. For instance, the Magic Leap One leverages a specialized spatial computing platform, designed to seamlessly integrate digital content into the user’s physical environment. Its display also automatically switches between two different focal planes based on sensed vergence information. In contrast, the HoloLens 2 employs a different combination of sensors and optics to generate a mixed reality experience. Meanwhile, the Meta Quest Pro relies on a more self-contained, immersive virtual reality experience, without some of the spatial awareness capabilities of the other two. Balancing these differences while ensuring that the studies were still rigorous and consistent demanded a meticulous approach.

To uphold the integrity and validity of the comparative analysis across all three platforms, it was imperative to carefully design tasks and procedures that could effectively harness the unique capabilities of each headset while maintaining methodological consistency. The user studies we performed were done in a way to at the same time leverage the unique strengths of each headset (not bring them down to the lowest common denominator) and to equitably yield

comparable, accurate and reliable eye tracking results. This involved meticulous planning, rigorous testing, and iterative refinement of the user study protocols.

An additional limitation lies in the tasks themselves. With our existing setup, there was no way to verify that participants performed the tasks correctly. A possible way to address this in the future would be to include a behavioral measurement to check the user's attention. For instance, this could involve periodically briefly changing the color of the tracking sphere and asking the user to click a clicker every time this occurs. Such a behavioral measurement would help verify that a user is paying attention to the tracking sphere and performing the task correctly, and data from users for whom this is not the case could be discarded. The absence of such a behavioral measure necessitates our assumption that the participants maintained their gaze on the target consistently and sincerely throughout the eye tracking task.

8 CONCLUSION

In this work, we compare the eye tracking capabilities of the Magic Leap One, HoloLens 2, and Meta Quest Pro during locomotion. The included tasks account for when a user is sitting with their head held still, sitting with their head freely moving, or walking, as well as whether a user is viewing screen-stabilized, body-stabilized, or world-stabilized content. We make the code for our test tasks and parameters available in an open-source distribution.

Our analysis revealed overall lower eye tracking accuracy while walking compared to stationary poses. Between the three tested headsets, we observed significant differences between the HoloLens 2 and the other headsets, especially while walking, which may be explained by different fit and tightness of the headset in front of the eyes. We demonstrated that we can enact improvements to eye tracking performance on the HoloLens 2 while walking when a calibration technique involving stipulated head motion is employed. Our work emphasizes the importance of considering eye tracking accuracy in different scenarios, rather than assuming stationary conditions.

Overall, our work provides future researchers with a means of investigating the accuracy of the eye tracker of their MR device prior to use. This may allow researchers to gain valuable insights, improving the performance of their eye tracker in mobile mixed reality applications.

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